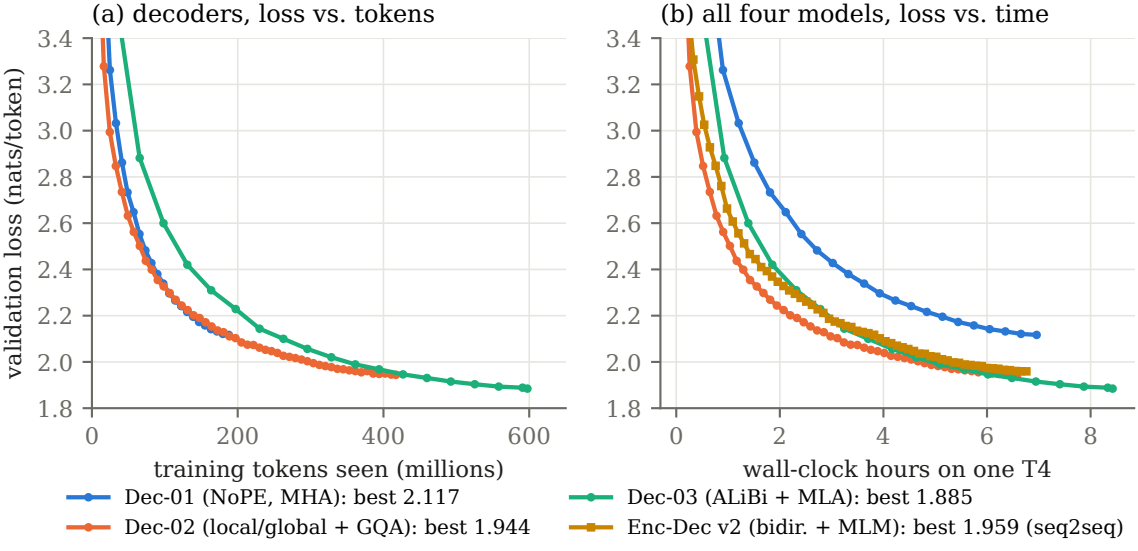
(a) decoders, loss vs. tokens
(b) all four models, loss vs. time
validation loss (nats/token)
3.4
3.2
3.0
2.8
2.6
2.4
2.2
2.0
1.8
0
200
400
600
training tokens seen (millions)
0
2
4
6
8
wall-clock hours on one T4
Dec-01 (NoPE, MHA): best 2.117
Dec-02 (local/global + GQA): best 1.944
Dec-03 (ALiBi + MLA): best 1.885
Enc-Dec v2 (bidir. + MLM): best 1.959 (seq2seq)